

TEST PLAN DOCUMENT

Project Name: Add Two Number Web App

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1.1 INTRODUCTION

This Test Plan document defines the testing strategy, objectives, scope, resources, schedule, and testing activities for the Add Two Numbers web application.

1.2 OBJECTIVE

The objective of testing is to verify that:

- The application performs addition correctly.
 - Frontend and backend communicate properly.
 - Data is stored correctly in MySQL database.
 - APIs work correctly.
 - Application behaves properly under valid and invalid conditions.
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1.3 SCOPE OF TESTING

In Scope

- UI testing
- Functional testing
- API testing
- Database testing
- Integration testing
- System testing

- Basic security testing

Out of Scope

- Load testing
- Cloud deployment testing
- Mobile application testing
- Cross-platform mobile testing

1.4 TEST ITEMS

The following components will be tested:

Component	Description
Frontend	HTML/CSS/JavaScript user interface
Backend API	FastAPI endpoints
Database	MySQL database
Integration	Frontend-backend-database flow

1.5 FEATURES TO BE TESTED

Feature ID	Feature Description
TP-001	Input fields validation
TP-002	Add button functionality
TP-003	Addition calculation
TP-004	Result display

Feature ID	Feature Description
TP-005	API response validation
TP-006	Database insertion
TP-007	Calculation history display

1.6 TESTING TYPES

Functional Testing

Verify all functionalities work correctly.

UI Testing

Verify layout, buttons, inputs, and responsiveness.

API Testing

Verify backend APIs return proper responses.

Database Testing

Verify data storage and retrieval.

Integration Testing

Verify frontend, backend, and database interaction.

System Testing

Verify complete workflow.

Regression Testing

Verify new changes do not break existing features.

1.7 TEST ENVIRONMENT

Item	Details
Operating System	Windows/macOS/Linux
Frontend	HTML, CSS, JavaScript
Backend	FastAPI
Database	MySQL
Browser	Chrome
API Tool	Postman
Automation Tool	Selenium

1.8 ENTRY CRITERIA

Testing will start when:

- Development completed.
 - Backend server running.
 - Database connected successfully.
 - Test environment ready.
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1.9 EXIT CRITERIA

Testing will stop when:

- All critical defects resolved.
- All major test cases passed.

- APIs functioning correctly.
- Database operations verified.

1.10 TEST SCENARIOS

Scenario ID	Scenario Description
TS-001	Verify webpage loads correctly
TS-002	Verify input fields visible
TS-003	Verify Add button functionality
TS-004	Verify addition of positive numbers
TS-005	Verify addition of negative numbers
TS-006	Verify decimal number addition
TS-007	Verify validation for empty input
TS-008	Verify API returns correct result
TS-009	Verify calculations stored in database
TS-010	Verify history displayed correctly
TS-011	Verify backend failure handling
TS-012	Verify invalid input handling

1.11 SAMPLE TEST CASES

Test Case 1

Field	Value
Test Case ID	TC-001

Field	Value
Scenario	Verify addition functionality
Steps	Enter 5 and 7 and click Add
Expected Result	Result should display 12
Actual Result	To be filled during execution
Status	Pass/Fail

Test Case 2

Field	Value
Test Case ID	TC-002
Scenario	Verify empty input validation
Steps	Leave inputs empty and click Add
Expected Result	Validation message displayed
Actual Result	To be filled during execution
Status	Pass/Fail

1.12 DEFECT MANAGEMENT

Defects found during testing should include:

- Defect ID
- Description
- Severity
- Priority

- Steps to reproduce
- Screenshot
- Status

1.13 RISKS AND MITIGATION

Risk	Mitigation
Database connection failure	Verify MySQL configuration
Backend server crash	Restart FastAPI server
Invalid API responses	Verify API logic
Incorrect calculations	Retest calculation module

1.14 AUTOMATION TESTING PLAN

UI Automation

Tool: Selenium

Automated Scenarios:

- Open webpage
- Enter values
- Click Add button
- Verify result

API Automation

Tool: pytest / requests

Automated Scenarios:

- Verify API response code
 - Verify JSON response
 - Verify invalid input handling
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1.15 DELIVERABLES

Testing deliverables include:

- Test Plan Document
 - Test Scenarios
 - Test Cases
 - Bug Reports
 - Test Execution Report
 - Automation Scripts
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CONCLUSION

The Add Two Numbers web application is a beginner-level full-stack project that demonstrates frontend development, backend API integration, MySQL database connectivity, and testing concepts. The project can be

used for learning software development, manual testing, automation testing, API testing, and database testing.